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Built Where It Is Used

Language, pedagogy, open technology and local partnership as conditions of effective and sustainable digital learning in Africa

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HIGHLIGHTS

- Localisation is modelled in three layers: language and content, pedagogy, and institutions (ownership, capability and data).
- Language mismatch is a documented barrier: in Côte d'Ivoire 55% of grade 5 pupils who spoke the test language at home reached basic reading, against 25% of those who did not.
- Kenyan trials that combined books in learners' languages with in-classroom teacher support improved reading.
- Imported models often adopt the form of reform without its function; open licensing and local partnership are practical means of building capability.
- Five tests of local substance and six testable propositions are offered for providers, funders and policy makers.

ABSTRACT

Purpose: Education reforms and technologies imported from elsewhere have a long record of looking successful while failing to change what happens in classrooms. This paper asks what it means for digital learning to be local, and whether locality bears on effectiveness and sustainability or only on sentiment. Design: A theory-building

narrative review draws on the literatures on design-reality gaps and isomorphic mimicry in development, language of instruction, learner-centred pedagogy, open educational resources and digital public goods, and on experimental evidence from Kenya. Findings: Locality operates at three layers. At the layer of language and content, mismatch between the language of instruction and the learner's language is a major, documented barrier, and Kenyan evidence shows that materials in the languages learners use, combined with in-classroom support, improve reading. At the layer of pedagogy, models promoted by international agencies have often been adopted in form without being sustained in practice, and teacher knowledge and support are binding constraints. At the layer of institutions, open licensing, exportable content, local hosting and local partnership determine whether a solution can be adapted, repaired and sustained by those who use it. Kenyan and continental policy frameworks give providers a clear mandate to build locally. Implications: Five tests of local substance and six testable propositions are offered. The paper also states the risks of the position, including the danger that locality becomes a slogan. Originality: The paper integrates development theory, language-in-education research and digital public goods standards into an operational framework for digital learning providers, and applies it reflexively.

Keywords: localisation; language of instruction; learner-centred pedagogy; open source; digital public goods; design-reality gaps; Kenya; Africa

1. Introduction

1.1 A recurring pattern

Development practice has a recurring pattern. A model that works in one place is exported to another, its visible features are adopted, and its results do not follow. Pritchett and Woolcock (2004) described how, in the 1960s and 1970s, developing countries copied developed-country models built on standardised, top-down programmes run by centralised civil services, an approach the authors judged to have failed in nearly every sector, and how later proposals for participatory approaches sparked unusual alliances and disputes. Heeks (2002) argued that information systems projects in developing countries fail, totally or partly, at high rates, and explained the pattern by the gap between the assumptions embedded in a design and the reality of the setting in which it is deployed. Education technology is a recent instance. Platforms, curricula and pedagogies cross borders because they are available, well funded and well marketed, and are then found not to fit the language, infrastructure, teachers or institutions that receive them.

The usual response is to say that technology should be local. The word is rarely defined, and it is easily taken for a matter of branding: a local name, a flag, a photograph of local learners.

This paper takes the claim more seriously. It asks whether locality has a measurable bearing on whether learners learn and whether solutions last, and what a provider would have to do to deserve the description.

1.2 The gap in the literature

Three bodies of work bear on the question, and they are seldom combined. The first is the development-theory literature on implementation failure, which gives concepts such as design-reality gaps and isomorphic mimicry but says little about education technology specifically (Heeks, 2002; Pritchett et al., 2013). The second is the education literature on language of instruction and on the transfer of pedagogical models, which is rich in evidence but rarely addresses digital platforms (Ouane & Glanz, 2011; Schweisfurth, 2013). The third is the literature and practice of open educational resources and digital public goods, which addresses ownership and sustainability but is rarely connected to evidence on learning (UNESCO, 2019; Digital Public Goods Alliance, n.d.). A provider who must decide what languages to build in, what pedagogy to adopt, what to license openly and whom to partner with needs the three together.

1.3 Contribution and structure

This paper makes three contributions. First, it distinguishes three layers of localisation, language and content, pedagogy, and institutions, and argues that effectiveness and sustainability depend more on the second and third than on the first. Second, it reviews evidence for each layer from Kenya, Sub-Saharan Africa and the wider literature, and derives five tests of local substance and six propositions. Third, it applies the tests reflexively to a Kenyan provider.

Section 2 describes the method. Section 3 develops the theoretical foundations and the layered framework. Sections 4 to 6 review the evidence at each layer. Section 7 reviews the policy frameworks that make the case for local provision. Section 8 proposes the five tests, Section 9 the propositions, and Section 10 applies the tests to Instracta. Sections 11 to 13 discuss implications, limitations and conclusions. The first author is the founder of the provider discussed in Section 10 and has an interest in the conclusions, which the closing statements declare.

2. Method

2.1 Type of review

This is a theory-building narrative review. It is not a systematic review conducted to a PRISMA protocol and reports no screening counts. It draws on heterogeneous literatures, including

development studies, comparative education, linguistics and information systems, in which a single protocol would be difficult to apply, and its purpose is to integrate them into a framework with testable propositions. The weakness of the approach is selection, and the steps below are intended to limit it.

2.2 Search, inclusion and appraisal

Searches were conducted in October 2026 using a general web search engine and repository records of publishers and research networks (including IDEAS/RePEc, EdTech Hub, Open Development and Education, the Global Education Monitoring Report and the UNESCO digital library). Strings combined localisation concepts with regional and design terms, for example language of instruction Kenya mother tongue, learner-centred pedagogy critique Africa, design-reality gaps developing countries, isomorphic mimicry, digital public goods standard, and open educational resources recommendation. Appendix A lists the principal strings.

Sources were included if they reported experimental or large-sample evidence from Kenya or Sub-Saharan Africa on language, pedagogy, teacher support or technology; made a substantive theoretical contribution to the study of implementation, pedagogy transfer or open resources; or established a policy or legal framework relevant to Kenyan providers. Policy and legal sources are summarised at a general level and should be checked against the primary text before being relied on in legal or policy settings. Vendor and promotional material was excluded. For several sources only an abstract or catalogue record could be accessed, and statements are limited to what those records report.

2.3 Positionality

The first author is a Kenyan practitioner: an instructional designer, learning management system lead at a Kenyan university and founder of a Kenyan provider. Local knowledge informed the questions asked and the practical tests proposed. The first author's interest in local provision is declared, and the paper includes an explicit statement of the risks and limits of the position (Section 12).

3. Theoretical foundations and the layered framework

3.1 Design-reality gaps and isomorphic mimicry

Heeks (2002) proposed that the success or failure of an information system in a developing country depends on the size of the gap between the design and the reality of its context. He identified three high-risk gaps: country context gaps, where the design does not fit the national setting; hard-soft gaps, where technical and social or human elements are misaligned; and private-public gaps, where the expectations of private-sector designs clash with public-sector

realities. He suggested that gaps could be narrowed through local improvisation. The framework transfers readily to learning platforms. A system designed for high-bandwidth, English-speaking, individually owned devices, deployed to learners on shared handsets and intermittent connections who speak other languages, presents several gaps at once.

Pritchett, Woolcock and Andrews (2013) described the related pattern of isomorphic mimicry: the adoption of the forms of functional institutions without their functions, so that a system looks like a successful one while performing no better. Andrews, Pritchett and Woolcock (2017) developed the argument that capability is built by locally driven problem-solving and iterative adaptation and not by transplanting best practice. For an education provider the lesson is to resist the supposition that a model validated elsewhere is validated here, and to treat adaptation as a core activity and not as a localisation afterthought.

3.2 Learning as situated and culturally relevant

Learning theory gives a second reason for fit. Brown, Collins and Duguid (1989) argued that knowledge is situated, being in part a product of the activity, context and culture in which it is developed and used, and that learning decontextualised from authentic activity is limited in its transfer. Ladson-Billings (1995) proposed that culturally relevant pedagogy requires that students experience academic success, develop cultural competence and develop a critical consciousness about the social order. Her work concerned African American students in the United States and its transfer is by analogy, but the principle that examples, cases and assessments should connect with learners' lives is consistent with the argument from situated cognition. Ngũgĩ wa Thiong'o (1986) made the broader argument that language carries a worldview and that education in a colonial language can separate learners from the community it is meant to serve.

Toyama (2015), reflecting on technology in development, argued that technology amplifies existing human capacities and institutions and does not substitute for them. Applied to learning, the claim implies that a platform will amplify the quality of the instruction, the capability of the teachers and the fit of the content that it finds, and that localisation is therefore not a cosmetic layer but part of what determines what is amplified.

3.3 The three layers

We distinguish three layers of localisation, shown in Figure 1. They differ in how visible they are and how hard they are to change.

1. Language and content. Whether the learner can understand the medium of instruction, and whether examples, cases and assessments reflect the learner's world.
2. Pedagogy. Whether the way of teaching suits the classroom, the teacher's training and the learner's expectations, or has been adopted because a funder or reform agenda required it.

3. Institutions. Who owns the platform and the content, who can host, adapt and repair it, where the data are held, and who is accountable to learners.

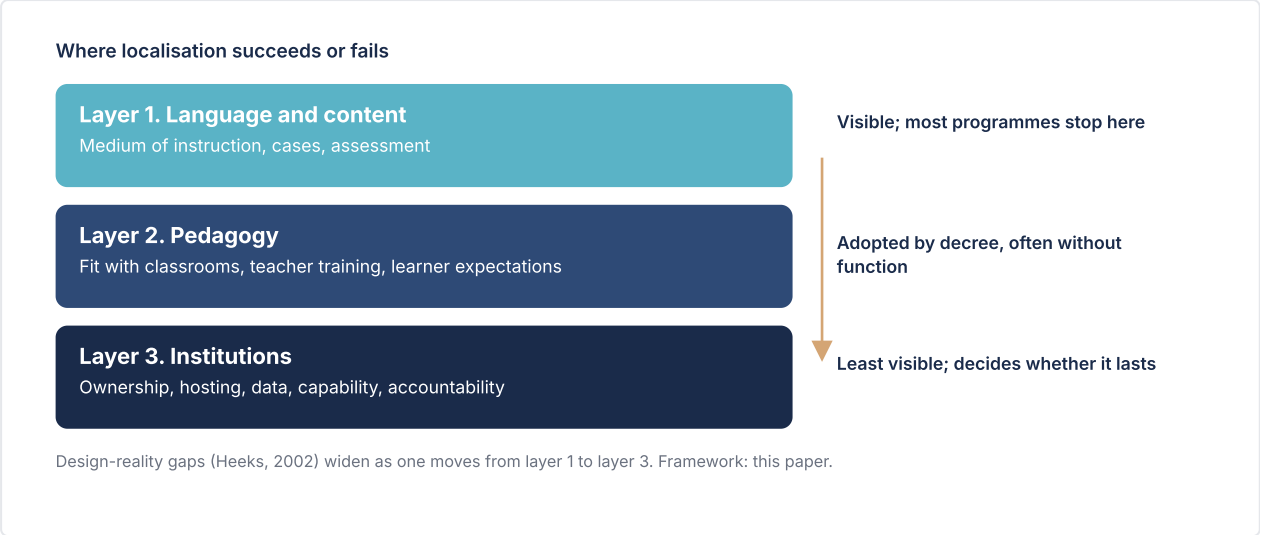


Figure 1. Three layers of localisation. Most programmes that claim to be local address the first layer, partially. Effectiveness and sustainability depend more on pedagogy and institutions, where design-reality gaps (Heeks, 2002) are largest and least visible.

The layers are not independent. A programme in a local language that rests on an imported pedagogy and a foreign-hosted proprietary platform is local at the surface and dependent at the core. Conversely, a programme with strong local ownership may still fail if its language excludes the learners it targets. The framework is a diagnostic: it asks, for any programme, which layers have been addressed and which are merely asserted.

3.4 Relation to the companion papers

This paper is the third of three working papers. The first argues that access should be assessed at four gates, the third of which, use, depends on language and design. The second proposes a proportionate evidence standard, which is the means by which the claims made here can be tested. The three are intended to be read together. Local provision without evidence is sentiment, and evidence without local fit may not travel.

4. Layer 1: language and content

4.1 The evidence on language of instruction

The evidence that language of instruction affects learning is among the strongest in comparative education. The 2016 UNESCO policy paper, If you don't understand, how can you

learn?, estimated that 40% of the world's population does not have access to education in a language they speak or understand, and illustrated the consequence with data from Côte d'Ivoire, where 55% of grade 5 students who spoke the language of the test at home reached basic reading in 2008, against 25% of those who did not (UNESCO, 2016). Figure 2 shows the comparison. Ouane and Glanz (2011), in an analytical review for the UNESCO Institute for Lifelong Learning and the Association for the Development of Education in Africa, examined the theory and practice of mother-tongue and bilingual education in sub-Saharan Africa, including costs, classroom practice and the role of locally based multilingual publishing, and argued that language choices should be addressed across policy, costing, governance, classroom practice, literacy and publishing.

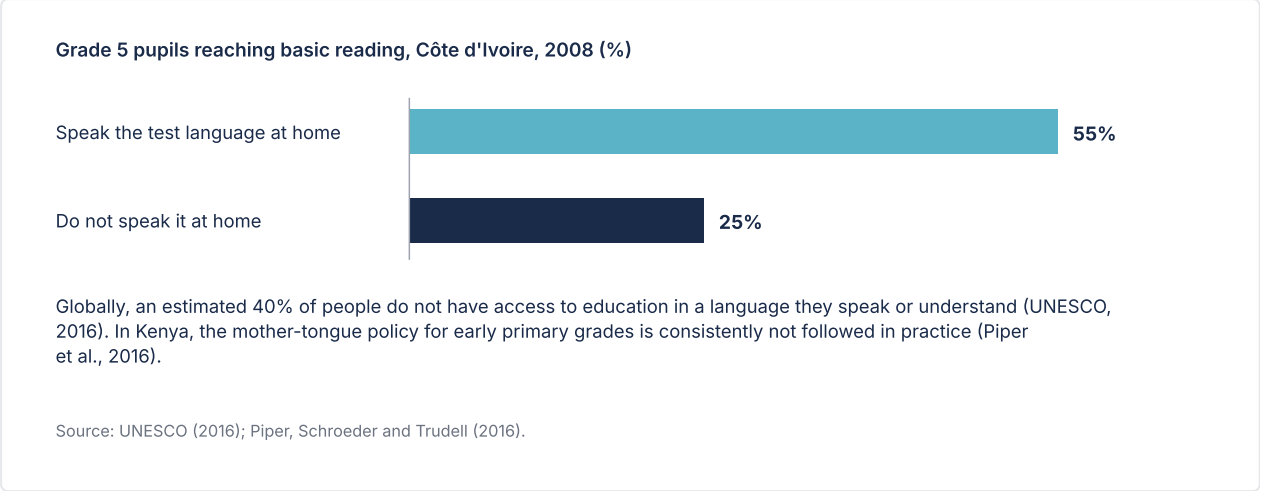


Figure 2. Grade 5 pupils reaching basic reading in Côte d'Ivoire, 2008, by whether they spoke the test language at home. Source: UNESCO (2016). The Côte d'Ivoire figures are an illustration of the pattern, not an estimate for Kenya.

The argument is not only that mother-tongue instruction is better. It is that language mismatch is a cost that learners bear silently, and that providers can reduce. Brock-Utne (2012) argued that the dominance of former colonial languages in African education reproduces inequality, as those who enter school speaking the language of instruction have an advantage over those who do not. The pattern relates directly to the first paper's third gate: a service built in a language the learner does not use fluently fails at the use gate whatever its price.

4.2 The Kenyan evidence

Kenya's constitution establishes Kiswahili as the national language, Kiswahili and English as the official languages, and an obligation on the state to promote and protect the diversity of the languages of the people (Republic of Kenya, 2010, Art. 7). The Basic Education Act of 2013 and the competency-based curriculum operate within that framework (Republic of Kenya, 2013). The practical reality is more complicated. Piper, Schroeder and Trudell (2016) examined oral

reading fluency and comprehension among about 2,000 Kenyan children tested in English, Kiswahili and either Dholuo or Gikuyu, their mother tongue. Children read English words more easily than Kiswahili or mother-tongue words, but their comprehension was significantly lower in English than in the other languages. The authors concluded that focusing on English fluency is an inefficient way to build comprehension, because pupils reach only minimal English fluency and modest comprehension in their own languages, and they found that Kenya's national policy of using the mother tongue as the medium of instruction in early primary grades is consistently not followed in practice.

The randomised evaluation of the PRIMR Initiative, in more than 400 schools in three counties, supplied books in English and Kiswahili alongside ongoing instructional support and classroom observation, and found improved oral reading fluency in both languages (Piper et al., 2014). The finding does not show that any one language is best. It shows that a package built around the languages in use, with support in the classroom, can move reading outcomes at scale. It also suggests that language and teacher support are complements.

4.3 Implications for digital providers: which languages, and for what

For a digital provider the practical question is which languages to build in and for which purposes. The evidence is mostly from early primary reading and does not settle the question for adult and workforce learners, who may have made a different adjustment between languages. Several principles follow. Providers should find out which languages their target learners use in learning and at work, and should not assume that English or Kiswahili suffices. They should make instructions, navigation and assessment instructions available in the languages learners use even where content is in English, since misunderstanding the task is a different problem from misunderstanding the content. They should work with language editors and subject experts who live in the context and should test comprehension with the learners for whom the material is intended. And they should publish the list of languages in which materials are available, so that funders and learners can see what is promised.

4.4 Content: whose examples, whose cases

Content is the less-studied half of this layer. The argument from situated cognition and culturally relevant pedagogy (Section 3.2) suggests that examples, cases and assessments should connect with learners' lives, work and communities. In practice this means using local case material, local data and local legal and policy references, and designing practice tasks in contexts that learners recognise. It also means avoiding the inverse error of treating local as the same as simple. A local case can be analytically demanding. The test of local content is that learners recognise the situation, and the test of quality is that the material is accurate, current and rigorous.

4.5 Multilingual design in practice

Few providers can translate all content into every relevant language, and the evidence does not require it. A defensible approach, which follows from the findings of Piper et al. (2016) that comprehension is lower in English even when decoding is easier, is to layer languages by function. The interface, navigation, error messages and assessment instructions come first, because misunderstanding the task is a cheaper problem to fix than misunderstanding the content and affects every learner. Glossaries of key terms in the languages learners use come next, followed by audio explanations of difficult concepts, which also support learners with low literacy or visual impairment and thus link to the access framework of the first paper. Full translation of core content is reserved for the programmes and languages where learner demand and sponsor funding justify the cost.

Three cautions apply. Machine translation can help but must be reviewed by speakers who know the subject, because errors in technical terms can mislead. Language use among Kenyan adults is often mixed, with code-switching between English, Kiswahili and mother tongues, and a provider should observe how its learners actually talk about the subject before deciding what to translate. And the decision should be evidence-led: a short comprehension check administered in two language versions to a sample of learners costs little and can show whether the translation is worth extending. This approach follows from the evidence reviewed but has not itself been tested for adult learners in Kenya, and is offered as a hypothesis (P1).

5. Layer 2: pedagogy

5.1 The limits of importing a way of teaching

Learner-centred education is the clearest example of a pedagogy promoted internationally with limited regard for fit. Schweisfurth (2013), in a book-length review, asked whether the approach suits every society and classroom, noting that supporters tie it to cognitive psychology and to the aims of democratic citizenship and a skilled workforce, while critics question whether it fits every cultural and resource context and point to persistent implementation problems in the Global South. Tabulawa (2003) argued that the appeal of learner-centred pedagogy to international aid agencies was mainly political and ideological and that it was promoted as a worldview intended to develop a preferred kind of society, although usually justified in cognitive and educational terms. These are contested claims, and neither author argues that learner-centred methods lack value. Their point is that adoption by decree, without the conditions that make a method workable, produces form without substance, which is what Pritchett et al. (2013) call isomorphic mimicry.

The relevance to digital learning is that many platforms embed a pedagogy: self-paced modules, discussion forums, peer assessment, badges. These are not neutral. They assume learners who are comfortable with self-direction, who have time and connectivity to participate in forums, and who value peer feedback. Where those assumptions do not hold, as in many

classrooms and training settings in Kenya, a platform's embedded pedagogy can lead to low completion that the provider misreads as a motivation problem.

5.2 What the evidence supports across contexts

The experimental evidence reviewed in the second paper is broadly consistent across settings on two points: targeting instruction to the learner's level, and supporting teachers over time. Banerjee et al. (2007), Duflo et al. (2011) and Muralidharan et al. (2019) found gains from instruction matched to learners' levels in India and Kenya, and Piper et al. (2014; 2018) and Conn (2017) found gains from in-classroom teacher support in Kenya and across Sub-Saharan Africa. These are not culturally specific pedagogies. They are design principles that appear to travel because they address a feature of many classrooms: wide variation in learner preparation and limited teacher support. Beyond those principles, the safest course is to test locally and adapt, as Andrews et al. (2017) recommend.

5.3 Teachers as the point where pedagogy meets context

Bold and colleagues (2017), using nationally representative data from seven Sub-Saharan African countries covering close to 40% of the region's population, found that students received about two hours and fifty minutes of teaching per day, just over half of scheduled time, and that teacher knowledge of content and pedagogy was limited and strongly associated with student achievement. Hennessy and colleagues (2022) reviewed technology-mediated teacher professional development in low- and middle-income countries and noted that professional development there is often limited and short-lived. A locally effective programme is therefore one that builds the capacity of local teachers and trainers and does not bypass them. The Kenyan trials, which paired materials with in-classroom coaching and observation, support the same conclusion (Piper et al., 2014; Piper et al., 2018).

For a digital provider, the implication is that facilitator training is part of the product. A course sold to a college or training centre should include not only content but also guidance and development for the people who will run it, including how to adapt it to their learners. Programmes that treat the facilitator as a delivery channel for the provider's pedagogy are likely to meet the resistance that Tabulawa and Schweisfurth describe.

5.4 Auditing the pedagogy a platform embeds

Because platforms embed pedagogical assumptions, a provider should audit them. Table 1 lists common features, the assumption each makes about learners and context, and the question a provider should ask before adopting it. The audit is a reasoned checklist and not an empirical result: its value is to make assumptions explicit so that they can be tested in a pilot.

Table 1. Pedagogical assumptions embedded in common platform features

Feature	Embedded assumption	Question to ask before adopting
Self-paced modules	Learners have time, privacy and self-regulation skills	Do our learners study in short, interrupted sessions on shared devices?
Discussion forums	Learners are comfortable writing in public and have time and data to return	Would a short voice or SMS prompt reach more learners?
Peer assessment	Learners trust and are trusted by peers and accept peer authority	Is peer feedback culturally acceptable here, and with what training?
Badges and leaderboards	Competition motivates and visible ranking is welcome	Could ranking discourage women or learners with low prior attainment?
Video-first content	Bandwidth, devices and sound are available	What is the data cost of a module and can it be downloaded once?
Automated quizzes	Items are in a language and register the learner uses	Were items tested for comprehension with local learners?

A provider that completes such an audit before launch is better placed to adapt, and its pilot becomes a test of specific assumptions and not a general trial of the product. The audit also connects the layers: a feature that fails the test of fit often fails the access gate of the first paper as well.

6. Layer 3: institutions, ownership and data

6.1 Why ownership decides sustainability

The third layer is the least visible and the most consequential for sustainability. A programme that depends on a foreign platform, a proprietary licence or an overseas server can be switched off by decisions that local institutions do not control. The Global Education Monitoring

Report of 2023 framed the issue as one of terms: whose interests technology serves, who frames the problems it is meant to solve, and whether it is appropriate, equitable, evidence-based and sustainable. Among its recommendations to governments were independent evaluation of education technology, pilot projects that reflect total cost of ownership, and transparency in spending and contracts with private companies (UNESCO, 2023).

Kenya has a notable precedent for locally originated technology that scaled. Suri and Jack (2016) estimated that access to the M-PESA mobile money system raised per capita consumption and lifted about 194,000 households, 2% of Kenyan households, out of poverty, with larger effects for female-headed households. The relevance is not that learning platforms will have the same effect. It is that a service built around how people in Kenya actually transact, on the devices they actually own, can reach large numbers of low-income users. That is the principle of fit applied to technology.

6.2 Openness as a means of local capability

Ownership is addressed by openness. The Digital Public Goods Standard lists as baseline indicators open licensing, clear ownership, platform independence, documentation, a mechanism for extracting data, and adherence to privacy laws and to standards and best practices (Digital Public Goods Alliance, n.d.). The UNESCO Recommendation on Open Educational Resources, adopted unanimously by the 193 member states of UNESCO in November 2019, encourages the creation, adaptation and sharing of openly licensed educational materials (UNESCO, 2019). Wiley and Hilton (2018) argued that the permissions that open licences grant, to retain, reuse, revise, remix and redistribute, enable teaching practices that are not possible with conventionally licensed materials. Open learning platforms such as Moodle, described by Dougiamas and Taylor (2003), allow institutions to host and modify the software that runs their courses.

The logic is practical. Open licences and open-source software allow an institution in Kenya to host, translate, adapt and repair what it uses without negotiating with a vendor, and allow capability to accumulate locally, in the staff who learn to run and modify the system. They do not guarantee quality or sustainability. Eghbal (2016) documented the dependence of open infrastructure on the often unpaid labour of maintainers and argued that its sustainability is a problem of funding and governance, not only of licences. An open-source programme without maintenance funding can be as fragile as a proprietary one. The test of openness is therefore not only the licence but whether the project has a maintenance plan and a community.

6.3 Data: sovereignty, privacy and trust

Data are a direct case. The Data Protection Act of 2019 gives individuals in Kenya rights over their personal data and places obligations on data controllers and processors, including the registration of some of them with the Office of the Data Protection Commissioner (Republic of Kenya, 2019). A learning platform that holds the records of Kenyan learners operates within

that regime. Providers that host data in-country where practicable, that can export a learner's records on request, that collect only what is needed and that state their data practices publicly are better placed to comply and to earn trust. The Digital Public Goods Standard's indicator on a mechanism for extracting data is the technical counterpart of the legal right to data portability: a learner or institution that can leave a platform with its content and records intact is less exposed to the provider's pricing or failure.

6.4 Local partnership

Ownership is also relational. A programme that is co-designed and co-delivered with Kenyan universities, training colleges, employers, community organisations and teachers draws on knowledge the provider does not have, builds capability outside the provider, and is accountable to people other than its funders. Partnerships with universities can supply research and evaluation capacity, which links this paper to the evidence standard in the second. Partnerships with employers can supply demand signals and placement opportunities. Partnerships with teachers and trainers can supply the facilitation the evidence favours. Figure 3 depicts the relationships. The aim is not to localise the provider's brand but to distribute capability, so that the programme can continue if the provider changes or disappears.

6.5 Total cost of ownership and the cost of exit

Institutions that adopt a learning platform take on costs beyond the licence: hosting, administration, training, support, customisation and the cost of leaving. The Global Education Monitoring Report recommends that pilot projects reflect total cost of ownership and that spending and contracts with private companies be transparent (UNESCO, 2023). For a platform, the cost of exit is the cost of moving courses, learner records and customisations to another system. Where content is held in proprietary formats, or where the provider controls the data, that cost can be prohibitive and the institution is effectively locked in. Open standards for content packaging, open licences and a documented data export reduce it. The practical test of ownership, in this sense, is whether an institution could move away from the provider in a reasonable time and at a reasonable cost, which is also an indicator of whether the provider's service has to earn its renewal.

Hosting location introduces a further trade-off. In-country hosting may help with latency, with some institutions' policies and with the perception of data sovereignty, but may cost more, and may offer less resilience and security than large international providers. The Data Protection Act requires attention to how personal data are processed and transferred (Republic of Kenya, 2019). A provider should decide hosting on a stated rationale, document it in a data processing statement, and give institutions a choice where feasible.

6.6 Sustaining an open provider

Openness raises the question of how a provider sustains itself. Several models are in use and can be combined. A provider can charge for services around open software, such as hosting, support, training and customisation, as many open-source companies do. It can seek grants to build and maintain open components as public goods, which is the model the Digital Public Goods community promotes and which aligns with the funding of Eghbal (2016)'s maintainers. It can charge institutions that can afford to pay and use the margin to underwrite learners who cannot, linking to the third requirement of the access paper. And it can build a community of institutions that share the cost of maintenance. Each model has a risk. Service revenue can crowd out investment in the public good; grant funding can be unstable; and communities can be slow to form. A provider that is open about its model, and that reports how much of its revenue and effort supports open components, gives funders a basis for deciding whether to support the public-good part separately.

7. Policy frameworks that make the case

Local provision is not only a preference of providers. It is where policy points. The African Union's Continental Education Strategy for Africa 2016 to 2025 sets out continental goals for education and training, and its Digital Transformation Strategy for Africa 2020 to 2030 sets a framework for digital development across the continent (African Union, 2016; African Union, 2020). Kenya's National Digital Masterplan 2022 to 2032, launched in April 2022, includes targets to extend fibre to 40,000 public schools and learning institutions, to create 25,000 hotspots for youth, to establish 1,450 digital innovation hubs, and to develop ICT skills and local software and manufacturing capacity, including two software factories expected to employ more than 10,000 software engineers (ITWeb Africa, 2022). The Constitution of Kenya recognises the right to education and the diversity of the languages of the people (Republic of Kenya, 2010).

These documents create demand for locally built and locally hosted digital learning and a standard against which providers can be asked to show how their work contributes. They also set a trap. Frameworks can be satisfied by statements and not by capability, which is the form-without-function that the development literature warns against (Pritchett et al., 2013). Providers who wish to be credible should therefore state specific, checkable commitments, such as the proportion of staff, partners and content creators based in the country, the languages in which materials are available, the share of intellectual property held openly, and where data are hosted. These are the subject of the tests in Section 8.

Public procurement is the place where the frameworks meet practice. Public universities, colleges and agencies in Kenya buy or commission learning platforms, courses and training. The criteria used in those purchases shape the market. Where procurement rewards low price and vendor reputation alone, it favours imported, proprietary offers, and local providers and open solutions struggle to compete. Where it also rewards open standards, data portability, local hosting where justified, facilitator development and evidence of effect, it favours the

properties that this paper argues matter for effectiveness and sustainability. The Global Education Monitoring Report recommends transparency in spending and contracts with private companies and the independent evaluation of education technology (UNESCO, 2023). The Digital Public Goods Standard offers a ready set of indicators (Digital Public Goods Alliance, n.d.). The authors are not aware of a published Kenyan procurement standard for digital learning that incorporates both, and its absence is a policy opportunity.

8. Five tests of whether a programme is local in substance

Drawing on the three layers, we propose five tests that a funder, partner or learner can apply to any digital learning programme. They are intended to be answerable with evidence and not with assurances. Table 2 states each test, the question it asks and the evidence to look for. Figure 3 shows the partnership relationships on which tests 3 and 5 depend.

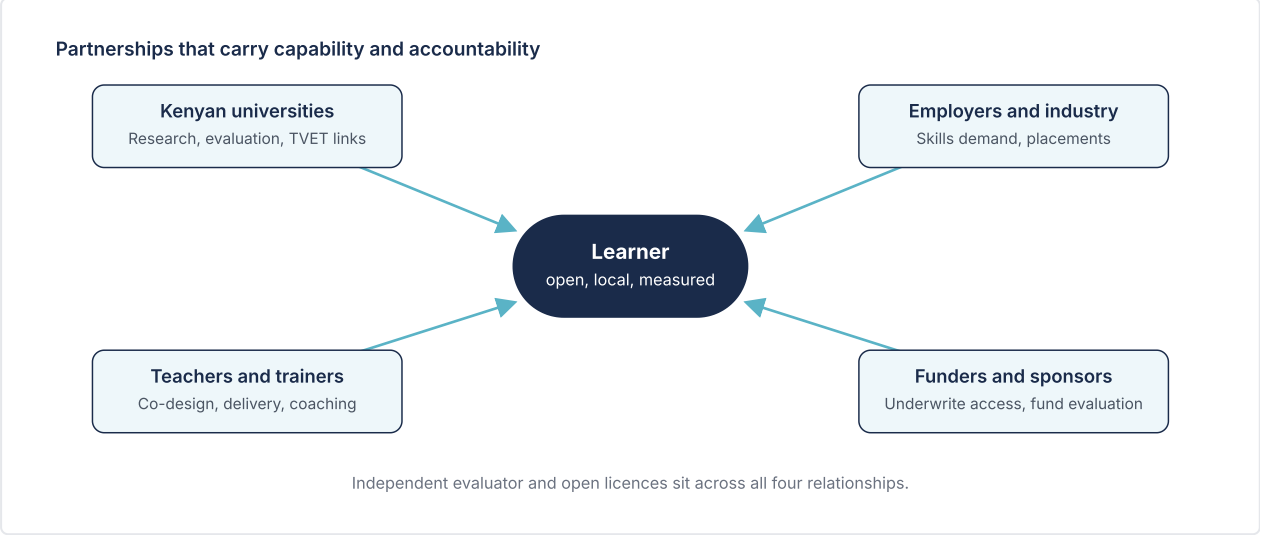


Figure 3. Local partnership model around the learner. Kenyan universities, employers, teachers and trainers, and funders each carry a different form of capability or accountability. An independent evaluator and open licences cut across all four relationships.

Table 2. Five tests of local substance

Test	Question	Evidence to look for
1. Language	Are materials and instructions available in the languages learners use, and have learners shown they understand them?	List of languages; comprehension results; use of local language editors
2. Fit	Was the pedagogy tested with local learners and teachers, and adapted as a result?	Pilot records; documented changes; facilitator feedback
3. People	Do local teachers, trainers and subject experts design and deliver, and does the programme build their capacity?	Staff and partner composition; training records; local evaluators
4. Ownership	Can a local institution host, adapt and exit the platform with its content and data intact?	Open licences; export formats; hosting arrangements; data processing statement
5. Accountability	Is the programme answerable to local learners and institutions, and is its evaluation independent?	Advisory structures; published results; independent evaluator

The tests are not a scoring system. They are a way of asking, of any programme, which layers have been addressed and which are merely asserted. A programme may pass some and fail others. The value lies in naming which. Tests 1 and 2 principally address the first two layers and the country-context and hard-soft gaps in Heeks' (2002) framework. Tests 3 to 5 principally address the institutional layer and the private-public gap, since they concern who delivers, who owns and who is answerable.

9. Propositions

The framework and evidence imply propositions that can be tested. They are stated as directional claims and are offered as a research agenda.

Table 3. Propositions derived from the layered framework

No.	Proposition	Layer and basis in the evidence reviewed
P1	Adult learners in Kenya will show higher comprehension and completion on materials whose instructions and key terms are available in the language they report using for study than on English-only materials of the same content.	Layer 1. UNESCO (2016); Piper et al. (2016); Ouane & Glanz (2011)
P2	Programmes that pilot with local learners and document adaptations will show smaller gaps between intended and observed use than programmes deployed without a pilot.	Layer 2. Heeks (2002); Andrews et al. (2017)
P3	Digital programmes accompanied by facilitator development will show larger learning gains than the same programmes delivered without it.	Layer 2. Piper et al. (2014; 2018); Conn (2017); Bold et al. (2017); Hennessy et al. (2022)
P4	Institutions that can host and adapt a platform will continue to use it after the provider's support ends at higher rates than institutions that depend on the provider for hosting.	Layer 3. Digital Public Goods Alliance (n.d.); Eghbal (2016)
P5	Open-source projects with funded maintenance will show lower rates of abandonment than those without.	Layer 3. Eghbal (2016); UNESCO (2019)
P6	Programmes with an independent local evaluator and a published results policy will report more null and negative findings than programmes without.	Layer 3. UNESCO (2023); Section 7 of the second paper

P1 to P6 are untested for Kenyan providers to the authors' knowledge.

10. Reflexive application: Instracta eLearning Solutions

10.1 Purpose and caution

This section applies the five tests to the provider with which the first author is associated. The exercise is reflexive and carries the risk of self-congratulation. It is therefore limited to stating

what the provider has in place, what it does not, and what it has committed to. It is not an evaluation.

10.2 Assessment against the tests

Instracta is a Kenyan provider based in Nairobi. It publishes its Moodle theme as open source and its course authoring tool produces content that can be exported and hosted elsewhere, which supports test 4. Its learning management system is in preparation and planned to be built on Moodle, an open-source platform, and its accessibility plugin is planned. It has not yet published a list of languages in which its materials are available (test 1), a data processing statement (test 4), or a record of local pilots with documented adaptations (test 2). It has proposed an independent evaluator and an advisory board, whose appointment is a condition of test 5, and it has not yet published a schedule of local partnerships with universities, training institutions and employers (test 3). Table 4 summarises the position.

Table 4. Provider status against the five tests (as at October 2026)

Test	In place	Not yet in place
1. Language	Awareness of the issue; English and Kiswahili capability	Published list of languages; comprehension testing
2. Fit	Instructional design practice grounded in Kenyan contexts	Documented pilots with adaptations
3. People	Kenyan team based in Nairobi	Published partnerships with universities, TVET and employers; facilitator development programme
4. Ownership	Open-source Moodle theme; exportable packages	Data processing statement; open licence policy for content; hosting arrangements documented
5. Accountability	Evaluator and advisory board proposed	Appointments confirmed; published results policy

10.3 What the exercise shows

As in the companion papers, the exercise produced a short, checkable list of gaps, most of them in documentation and publication and not in intention. Several are inexpensive to close: a list of languages, a data processing statement and an open licence policy can be published within weeks. Others, such as partnerships and facilitator development, need time and funding. The exercise also illustrates a risk in the framework. A provider can pass tests 4 and 5 on paper, by publishing licences and appointing a committee, without changing practice. The remedy is to ask for evidence of use: whether any institution has actually hosted or adapted the provider's software, and whether the evaluator has actually published a report that the provider did not edit.

11. Discussion and implications

11.1 For providers

For providers, the most useful result is a change in what counts as localisation. Translating the interface and adding local photographs is a first-layer activity and is the least costly to do and the least important to outcomes. The larger tasks are to fit the pedagogy to the setting, to develop the people who deliver it, and to design the platform and content so that a Kenyan institution can run and adapt them. Providers should publish their positions against the five tests and revisit them annually. They should also be honest about where foreign solutions are better than local ones and use them where they are.

11.2 For funders

For funders, the framework suggests due-diligence questions. In which languages are the materials available, and how was comprehension tested? What did the provider change after piloting? Who delivers the programme, and who builds their capacity? Can a Kenyan institution host and adapt the platform, and has one done so? Where are learner data held, and under what agreement? Who evaluates the programme, and are results published regardless of outcome? Funders also have a role in resourcing the less visible layers. Facilitator development, maintenance of open-source components and independent evaluation are rarely funded as such. A funder that pays only for delivery may be paying for the layer that matters least to sustainability.

11.3 For policy

For policy makers, the frameworks reviewed in Section 7 provide a mandate for local provision. Several instruments would strengthen it: procurement criteria that favour open licensing, data export and local hosting where practicable; support for the maintenance of open-source education software used in public institutions; recognition of facilitator development as part of any publicly funded digital learning programme; and support for the creation and sharing of

Kenyan-language and locally contextualised open educational resources, consistent with the 2019 UNESCO Recommendation (UNESCO, 2019). The Digital Public Goods Standard offers a checklist that Kenyan agencies could adapt for their own procurement (Digital Public Goods Alliance, n.d.).

11.4 For research

Three questions would most advance the field. First, does the availability of instructions and key terms in the learner's own language improve comprehension and completion among Kenyan adult learners, relative to English-only materials (P1)? Second, do institutions that host and adapt a platform sustain its use longer than those that depend on the provider (P4)? Third, what are the costs and benefits of the less visible layers, facilitator development and maintenance, in terms of learning gains per unit cost (P3 and P5)? Each can be tested in a pilot with a comparison group, and the results would be a public good for providers, funders and governments.

Two further research needs are methodological. First, measures of local substance are underdeveloped. The five tests proposed in this paper are qualitative and rely on documentary evidence and stakeholder testimony; validated instruments, tested for reliability across raters and settings, would allow comparison between programmes and over time. Second, the cost side of localisation is poorly documented. Few providers publish what it costs to translate, adapt, train facilitators, host locally and maintain open materials, and without such figures funders cannot compare the cost per learner of a local model with that of an imported one. Studies that report both costs and learning outcomes, using the cost-effectiveness conventions applied in education economics, would turn an argument that now rests on principle into one that can be judged on evidence. Replication across counties, languages and institution types would show how far the propositions hold beyond the settings in which they were developed, and where they fail. Qualitative work with learners, facilitators and institutional partners would add what trials cannot: an account of why adaptations are adopted or abandoned, and of how learners themselves judge whether a programme belongs to them. Such work should be designed with local researchers as full partners, and its findings returned to the communities studied.

11.5 Equity and the local: whose locality?

A final implication concerns who is included in the local. Kenya is multilingual and multicultural, and its learners differ by region, class, gender, disability and religion. A programme that is local to Nairobi may not be local to Turkana, to a coastal town or to a refugee settlement. Local design that listens only to the most accessible voices, for example urban, English-speaking, male, institutional, reproduces the exclusions that the first paper describes. The remedy is to treat locality as plural and to include, in design and piloting, learners at the margin: women, people with disabilities, rural and pastoralist learners, and speakers of languages with fewer

resources. Doing so is more expensive than designing for the centre, and it is the more reliable test of whether the claim to be local is substantive.

A related point concerns the knowledge that local partners bring. Partners are often treated as access channels, providing venues and recruiting learners, while design and decisions remain with the provider. Substantive partnership gives local partners a role in decisions about content, pedagogy, assessment and evaluation, and gives them a share of the credit and, where relevant, of the revenue. Ngũgĩ wa Thiong'o's (1986) argument about the politics of language is a reminder that who decides what counts as knowledge is itself a question of power, and the practical test is whether local partners can change what the programme does.

11.6 Sequencing localisation under budget constraints

Small providers cannot address every layer at once, and the framework should help them choose. We suggest three heuristics. First, address the institutional layer early, because decisions about licensing, hosting and data are cheap to make at the start and expensive to reverse once content and users have accumulated. A course built in an exportable format from the first day costs little more than one built in a closed format, while a course that must later be rebuilt for export costs as much as a second build. Second, invest in pedagogy where the evidence is strongest, which for the Kenyan trials reviewed above is in-classroom support for teachers and facilitators. A modest facilitator guide and a short training session may produce more learning than a large investment in new media. Third, treat language as a matter of progressive disclosure: begin with instructions, key terms and feedback in the learner's language, which are inexpensive, and extend to full instructional content where the evidence of need and the available resources justify it.

These heuristics are proposals and not findings. Their value is that they convert the framework from a description into a sequence of decisions, and each can be revisited in the light of pilot results. A provider that records which decisions it made at which layer, and why, creates a trail that funders and partners can inspect, and that supports the reflexive discipline the paper recommends. They also guard against a common failure, in which a provider spends its localisation budget on visible, first-layer changes because they are easy to show, while the second and third layers, where dependence is created or avoided, are left to chance.

Finally, sequencing should reflect the learner's perspective as well as the provider's. A learner meets the language first, the pedagogy second and the institution last, and often never sees the institution at all. The provider, by contrast, controls the institutional layer from the beginning and should therefore take responsibility for the layer that learners cannot see. The asymmetry is a further reason to publish decisions about licensing, hosting and data, so that learners, partners and funders can judge them on the provider's behalf.

11.7 An illustrative localisation audit of a single course

To show how the layers apply to a concrete product, consider a hypothetical ten-hour online course on workplace communication, built originally for an international audience in English with video lectures, a discussion forum and peer-assessed assignments. The example is constructed for illustration and does not describe any particular course.

At layer 1, the audit finds that the instructions and assessment items are in English only, that the case studies concern companies in other countries, and that the videos have no transcripts. The adaptation is to provide instructions and key terms in Kiswahili, to replace the cases with Kenyan workplaces, and to add transcripts and audio-only versions. At layer 2, the audit finds that the course assumes self-directed study over six weeks and that the forum has low participation in the pilot, since learners have little time and data. The adaptation is to cut the course into short units that can be completed on a handset in short sessions, to replace the forum with a weekly voice-note task shared in a small group, and to train a facilitator at each partner institution to run a weekly session. At layer 3, the audit finds that the course lives only on the provider's platform, with licensed media that cannot be redistributed. The adaptation is to export the course in a standard package that a partner college can host, to replace licensed media with openly licensed material or original content, and to publish a data processing statement.

Each adaptation is cheap relative to the initial build, and the sum changes what the course is. The audit also shows the value of the order in the framework: the layer 1 changes would have helped with comprehension but would not have fixed the forum or the platform dependence. The audit would be repeated after a pilot, with measures of comprehension, completion and learning gain, and the results published, linking the localisation work to the evidence standard of the second paper.

11.8 Counter-arguments and replies

Four counter-arguments deserve a reply. The first is that local provision fragments scale: a product adapted to each context cannot reach the numbers that a single global product can. The reply is that scale without fit produces enrolment without learning, and that open licensing allows adaptations to be shared, so that the cost of adapting is borne once and spread. The second is that international best practice is evidence-based and local variations are not. The reply is that the evidence for most best practice comes from other contexts, that effect sizes vary with setting (Vivalt, 2020; Pritchett & Sandefur, 2015), and that local testing is the means by which international evidence is checked. The third is that open source is unsustainable. The reply is that some projects are and some are not, that the problem is one of funding and governance (Eghbal, 2016), and that proprietary platforms also fail when providers change or withdraw. The fourth is that local hiring and partnership are slower and more costly. The reply is that they are, in the short run, and that the benefits accrue in capability and accountability that last beyond the programme. None of the replies is conclusive, and each is a candidate for empirical testing.

12. Risks, limitations and counter-arguments

12.1 Risks of the position

The argument for local provision carries risks that should be stated. Local is not the same as good: a locally built product can be poor, and foreign designs sometimes outperform local ones. Insistence on locality can slide into parochialism, or into essentialising a culture and treating learners as members of a type and not as individuals. Open licences can be adopted for appearance while the work of maintenance is neglected, a risk that Eghbal (2016) documents. Local partnerships can become tokenistic, with a local name on a programme whose design and decisions remain elsewhere. And locality can be used to resist legitimate comparison with international standards. The tests in Section 8 are designed to reduce these risks by asking for evidence. They do not eliminate them.

12.2 Limitations of the review

The paper has limitations that bear on its conclusions. It is a narrative review without a registered protocol and reports no screening counts, so selection bias cannot be excluded. The evidence that locality as such causes better learning is thin. What the literature supports is narrower: that language mismatch harms learning, that teacher support and instruction matched to the learner work across contexts, that imported reforms often fail to take hold, and that institutional ownership is a precondition for sustainability. The causal contribution of any single local feature is not established. Much of the Kenyan evidence concerns early-grade literacy and not adult and workforce learning. The Côte d'Ivoire data are an illustration from another country. The statements on Kenyan statutes and on continental strategies are summarised at a general level and should be checked against the primary text. The layered framework and the five tests have not been validated, and the reflexive assessment is a self-assessment by an interested party.

13. Conclusion

To say that digital learning should be local is to make a claim about effectiveness and sustainability, not about identity. The evidence supports it where language, pedagogy and institutions are concerned: learners learn more when they understand the language, teachers teach better when they are supported over time, imported models often fail to take hold, and institutions can only sustain what they can own and adapt. The claim is limited by the thinness of direct evidence on locality as such, and by the risk that locality becomes a slogan. The remedy is to turn it into tests that can be failed, and to publish the results. A provider that does so earns the trust that the word is meant to signal. The layered framework, the five tests and the six propositions offered here are intended to make that practice routine, and to make the claim of local provision something that can be checked.

Declarations

Declaration of competing interest

The first author is the founder of Instracta eLearning Solutions, whose products and partnerships are discussed in Section 10. No other competing interests are declared.

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Author contributions (CRediT)

Opiyo Michael Ouru: Conceptualization, Methodology, Investigation, Writing: original draft, Writing: review and editing.

Kenneth Goga Riany: Writing: review and editing.

Daisy Chepkoech O'maera: Writing: review and editing.

Data availability

No new data were generated. All sources are listed in the reference list; the search strings used are given in Appendix A.

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Appendix A. Search strings and source types

Searches were conducted in October 2026. Searches were iterative and results were screened by title and abstract against the criteria in Section 2.2.

Table A1. Principal search strings and source types

Layer	Search string (abridged)	Source types
Language and content	language of instruction Kenya mother tongue; if you don't understand how can you learn; Kenya oral reading fluency comprehension multilingual; optimising learning education publishing Africa language factor	UNESCO, RTI, Journal of Research in Reading, ADEA/UIL
Pedagogy	learner-centred education critique Africa; international aid agencies learner-centred pedagogy; PRIMR randomised Kenya; teacher professional development technology LMIC	Comparative Education, Routledge, IJED, World Development, Computers and Education Open
Institutions	digital public goods standard; open educational resources recommendation 2019; mobile money Kenya poverty; Data Protection Act Kenya 2019	DPGA, UNESCO, Science, Republic of Kenya
Theory and policy	design-reality gaps information systems developing countries; isomorphic mimicry state capability; Continental Education Strategy for Africa; Kenya National Digital Masterplan 2022-2032	The Information Society, Journal of Development Studies, AU, trade press

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